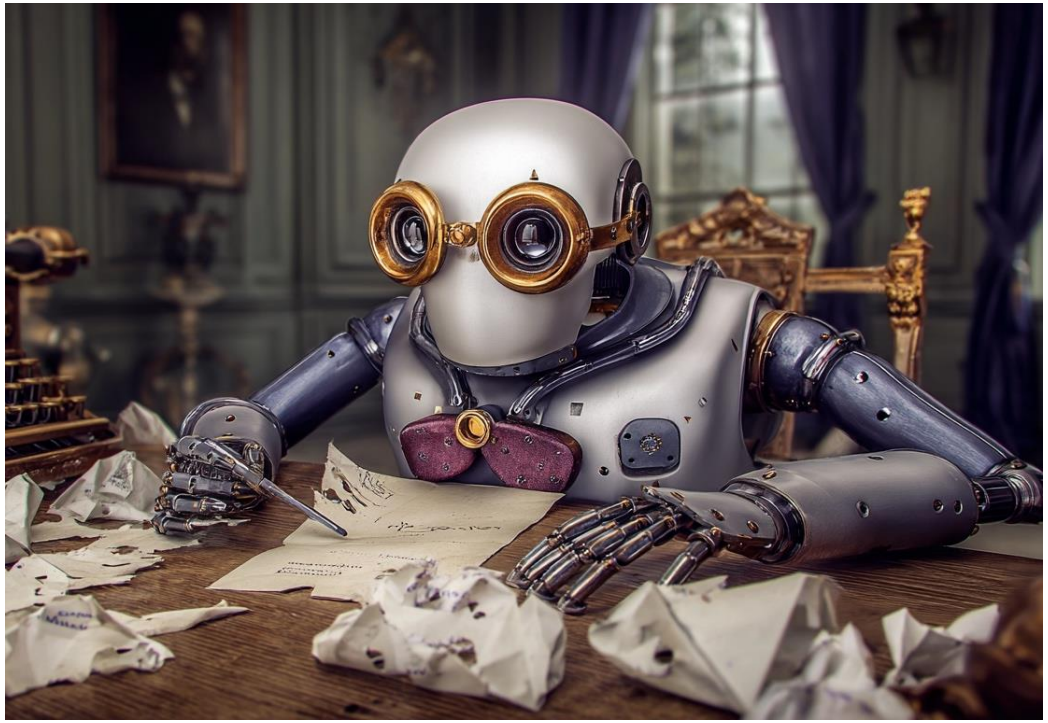
10 AI and Environment

GEB2508 AI and Society

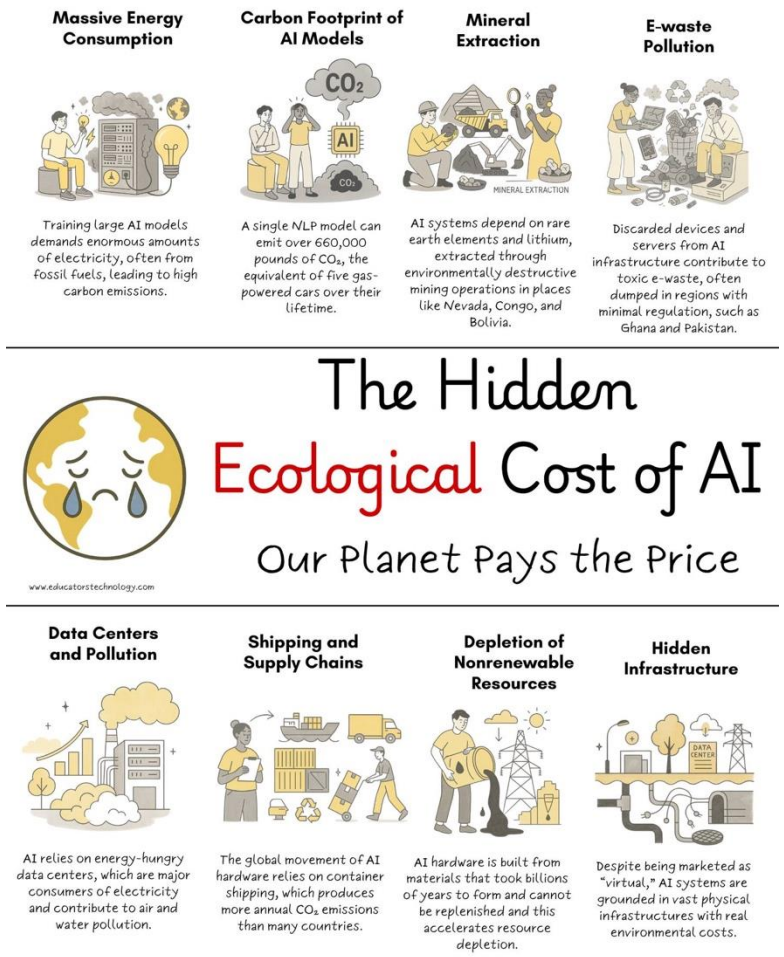
Review of Last Class

- Authorship and Copyright
  - Authorship in the Age of AI
  - Copyright and Training Data
- Political Economy of Knowledge
  - Knowledge as A Public Good
  - Enclosure of The Commons
  - Open-Source vs. Proprietary AI
- Knowledge Governance Regimes
  - Technocracy and AI



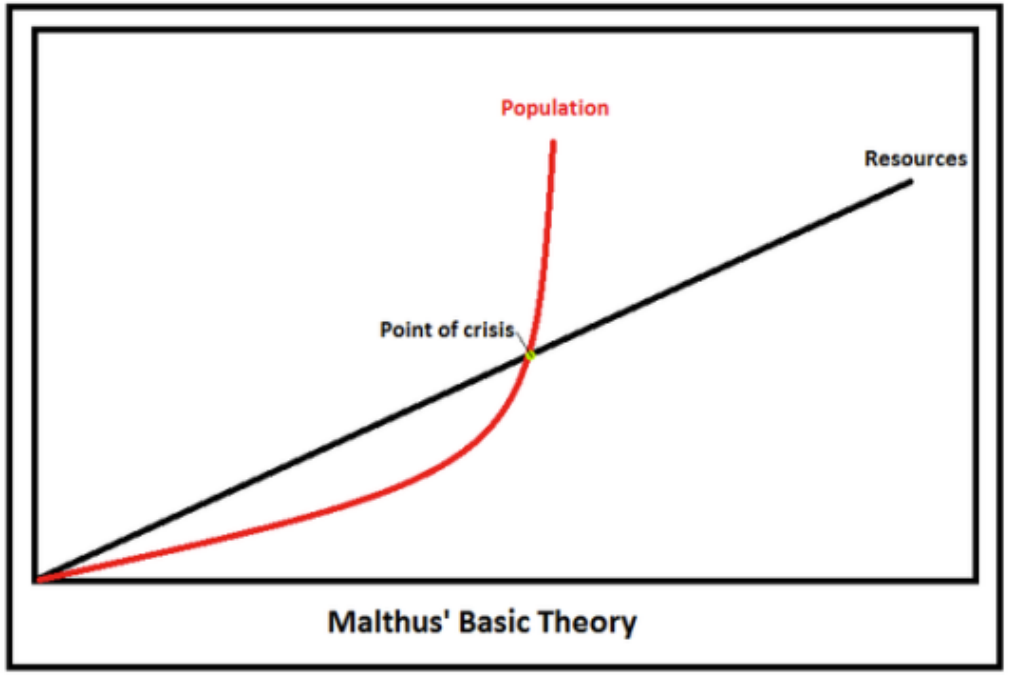
Content

- Technology-Environment Relations
  - Environmental Malthusianism vs. Cornucopianism
  - Environmental Cost of AI Infrastructure
  - Environmental Justice and Resource Politics of AI
- AI and Climate Change
  - Tragedy of The Global Carbon Commons
  - Energy Pathways for Powering AI
  - AI for Climate Science



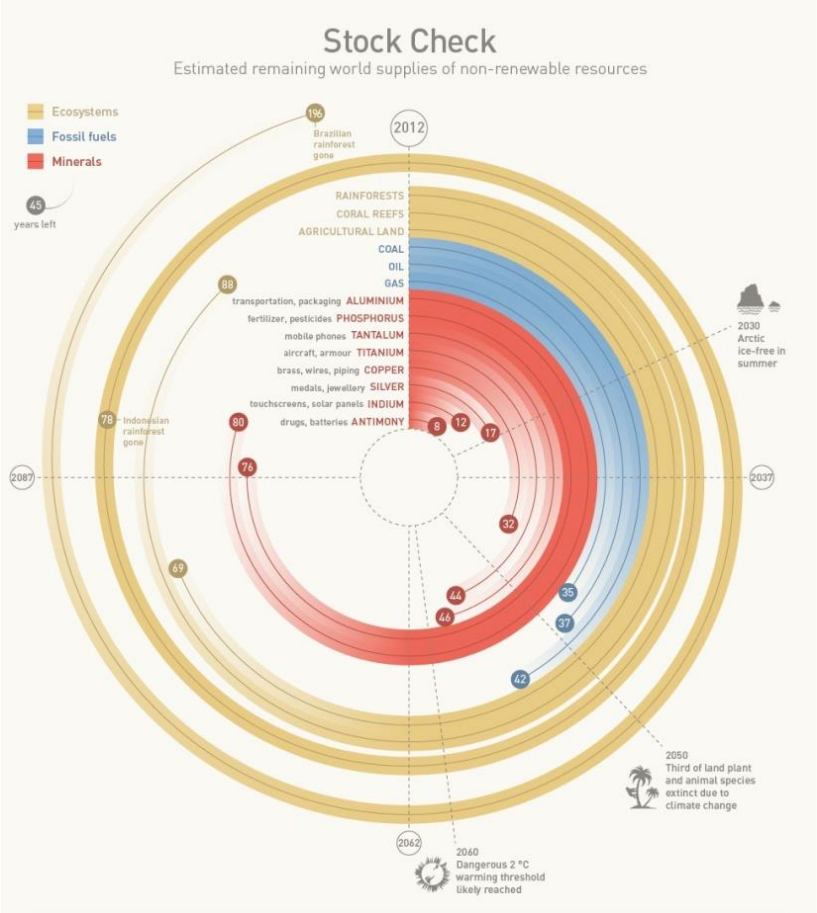
Malthusian Theory of Population

- Thomas Malthus (1766-1834)
  - *An Essay on the Principle of Population as It Affects the Future Improvement of Society* (1798)
  - Population growth will always **outgrow** the food supply.
- Preventive checks (birth control) and positive checks (war, disease, hunger, and famine)
  - Critique of welfare policies as ineffective and counterproductive



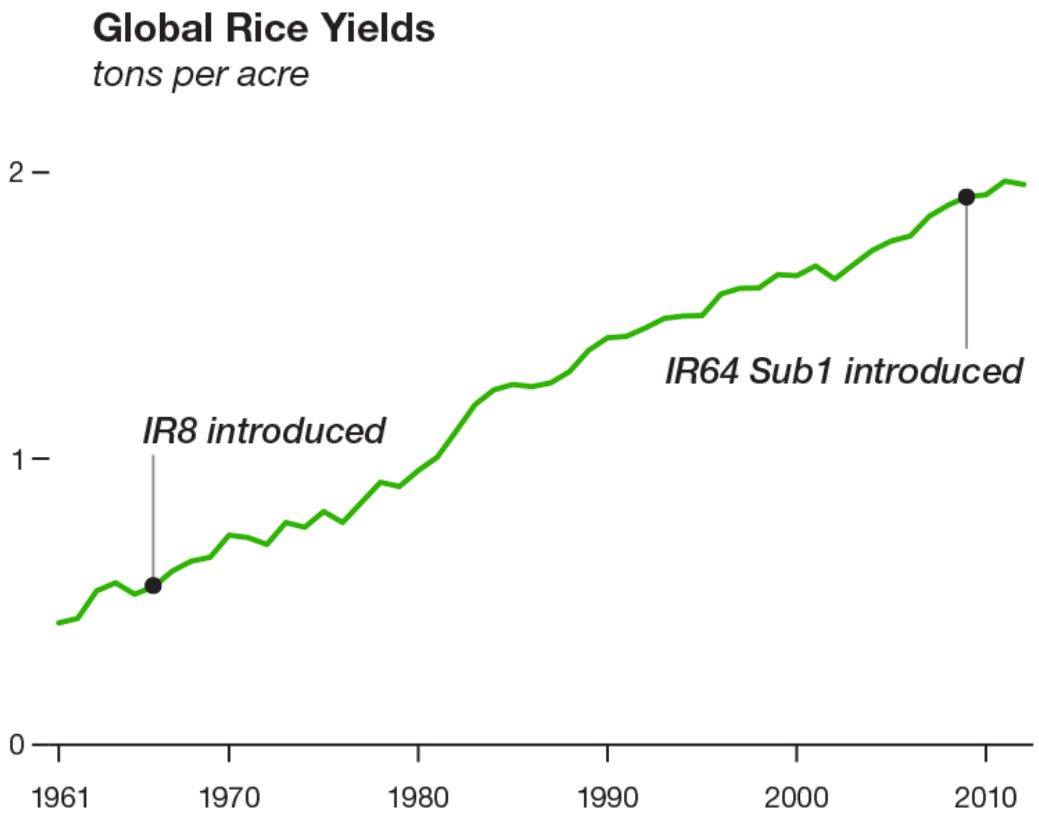
Environmental Malthusianism

- Neo-Malthusian thinking: from **food** to natural **resources**
  - *The Population Bomb* (1968)
  - *Limits to Growth* (1972)
- Criticisms: flawed assumptions and racist agendas; misplaced focus and unjust blame
- Reorientation of development policy toward **fertility control** and family planning



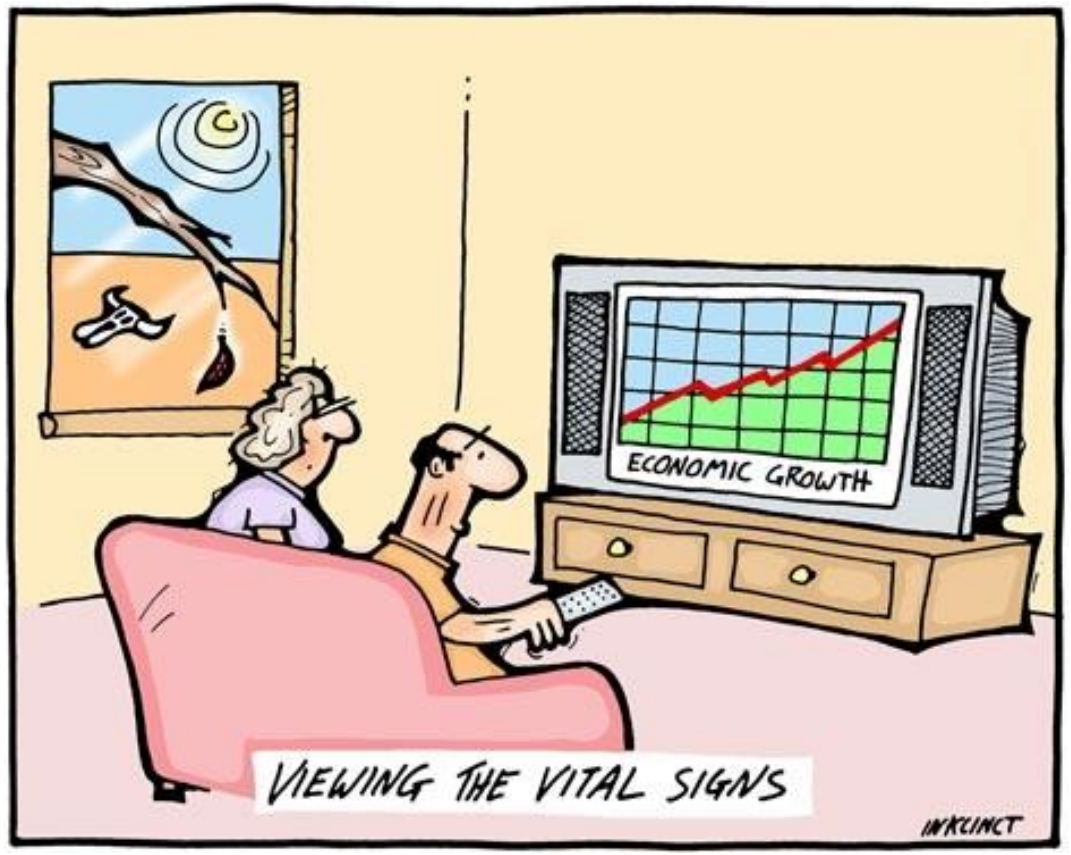
Green Revolution

- The productivity of food grains increased significantly due to **technological** advances.
  - Dr. Norman E. Borlaug - 1970 Nobel Peace Prize Laureate
- Positive impacts: increase in production, poverty reduction, decrease in costs of food grains
- Negative impacts: depletion of water, soil erosion, unemployment among farmers



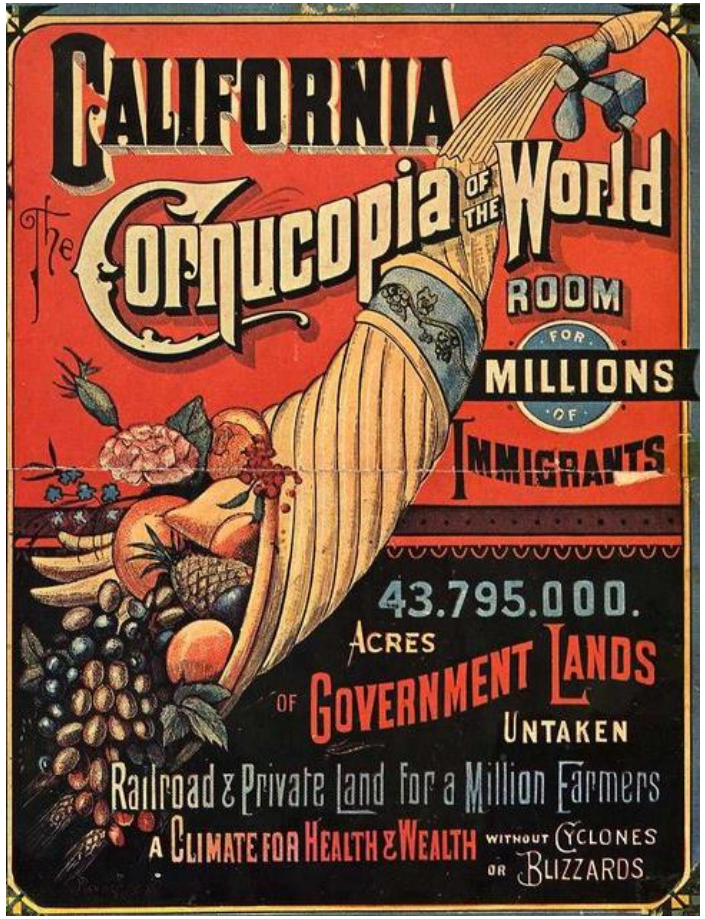
Technology and Environment

- Are Malthusian predictions proven wrong?
  - Technology has enabled humanity to exceed the limit.
- Technological developments have **expanded** and **accelerated** the processes of resource depletion and environmental degradation.
  - Resource depletion, pollution, climate change, biodiversity loss



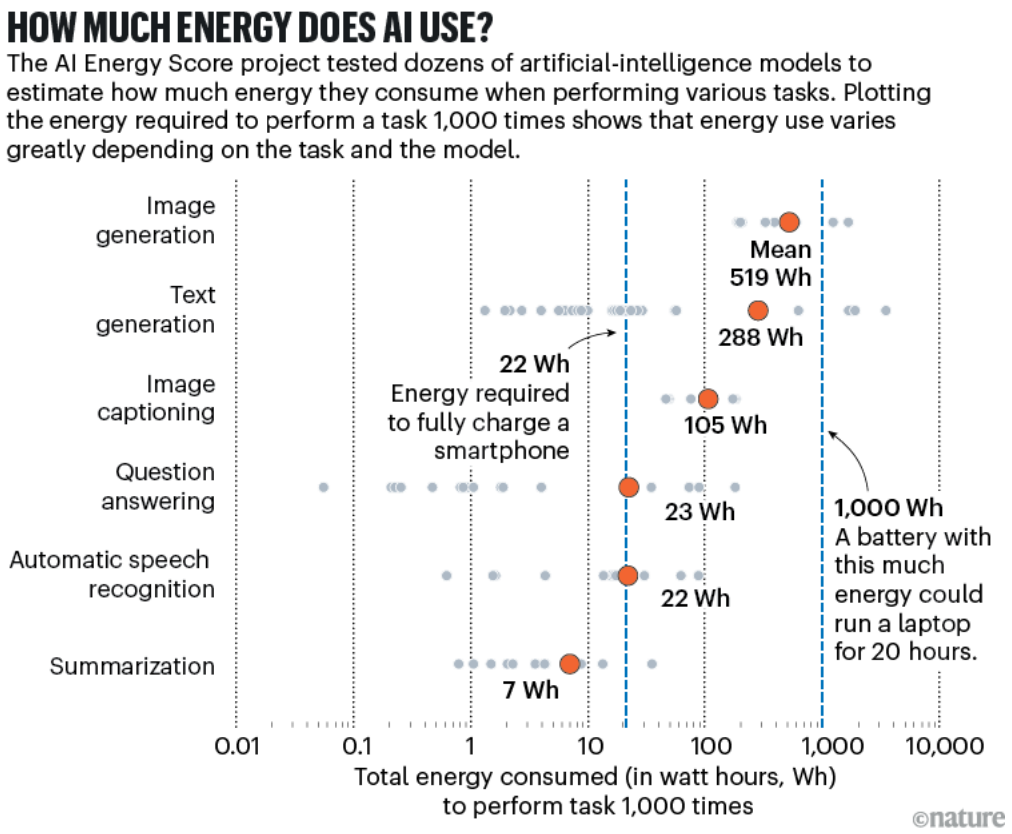
Cornucopianism

- Cornucopia: (Greek.) the horn of plenty
- A cornucopian is someone who believes the world's environmental problems don't exist or can always be **fixed by technology** or the free market.
- Technological optimism
  - E.g., alternative energy sources, outer space, etc.



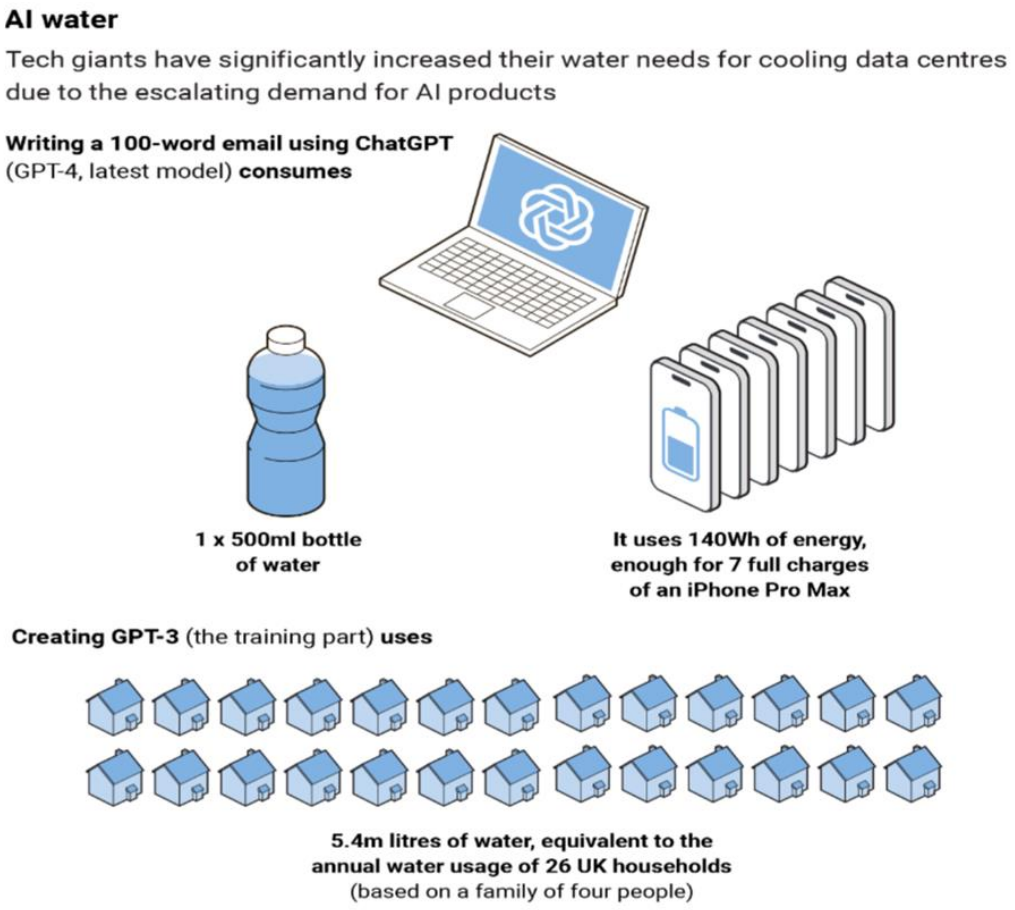
Environmental Footprint of AI: Power

- AI as a major driver of electricity demand growth
  - **Training** energy use: intense
  - **Inference** energy use: long-term
- Energy use concentrated in hyperscale data centers
  - ~1–2% of global electricity use
  - Data center siting favors **cheap electricity** and available land
  - Few jobs vs. significant energy demands



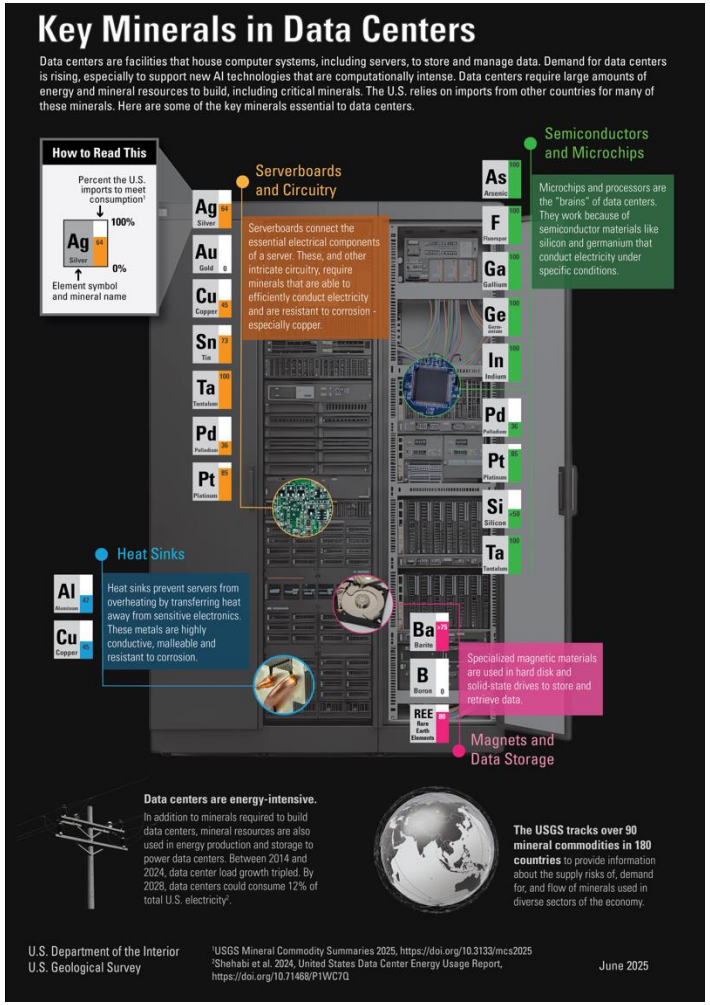
Environmental Footprint of AI: Water

- Lifecycle dependence: chips → model training → everyday inference
- **Evaporative cooling** for energy-dense data centers & water-intensive electricity production
- **Place-based** water issues and **justice-oriented** stakes
- Peak AI demand → peak cooling demand during hot weather



Extraction and Environmental Justice

- AI as part of planetary resource extraction
- **Lifecycle** displacement: upstream mineral extraction → manufacturing → downstream e-waste disposal
- Environmental justice: ecological burdens fall disproportionately on the Global South
  - Export of e-waste and **informal** recycling economies



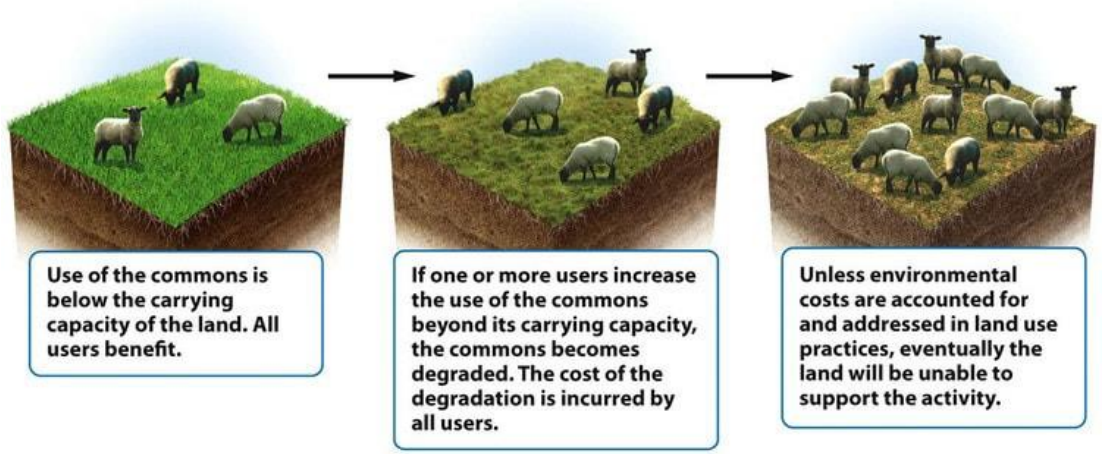
Extractive Ecologies of AI

Lithium mining in Latin America

Cobalt extraction in Congo

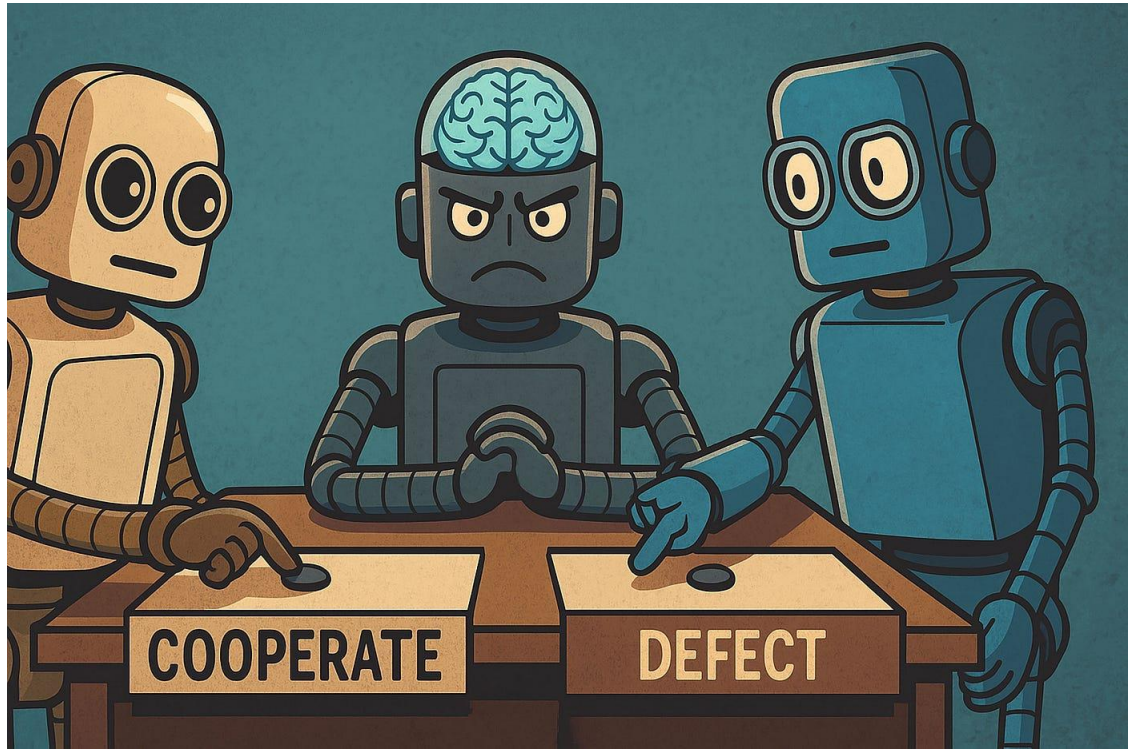
Tragedy of the Commons

- Hardin (1968) argued that the main difficulty in solving environmental problems lies in the conflict between individuals' **short-term** interests and society's long-term welfare.
- Such failure to cooperate around environmental problems often results in the degradation of collective goods.



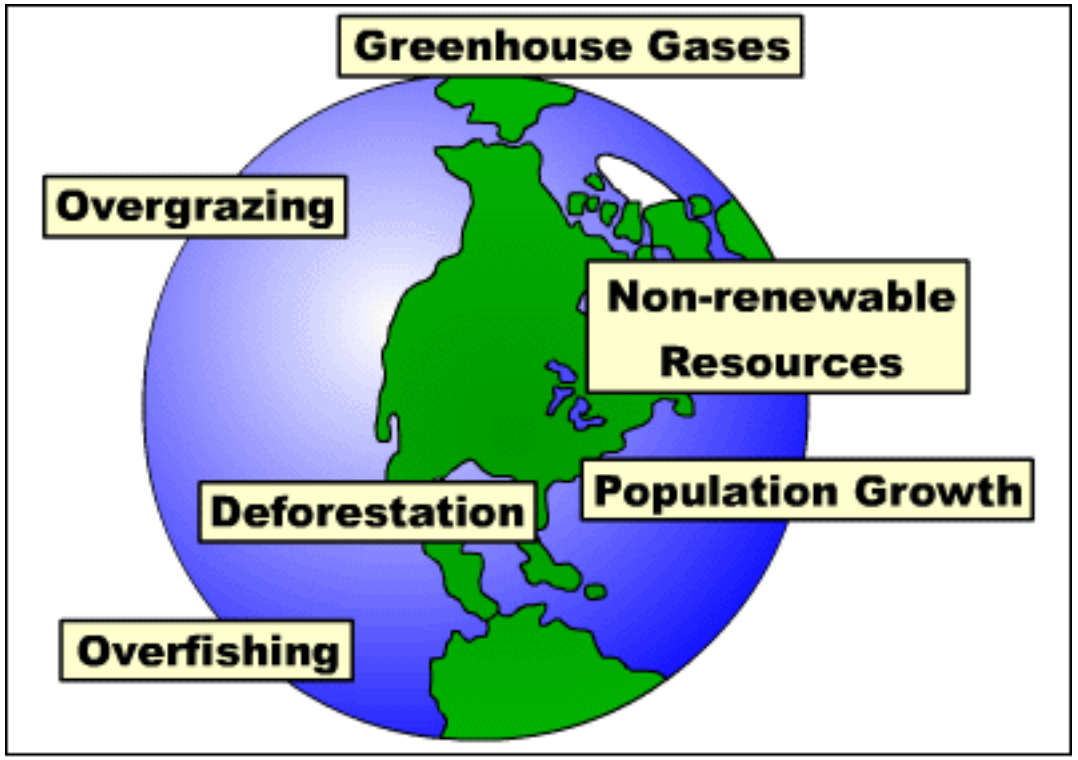
The Climate Prisoner’s Dilemma

- Prisoner’s dilemma: each player **rationally** defects to maximize individual payoff → mutual defection leaves both worse off than cooperation
- Climate prisoner’s dilemma: each country **rationally** continues emitting to sustain economic growth → collective over-emission produces climate change that ultimately harms all



Tragedy of the Global Carbon Commons

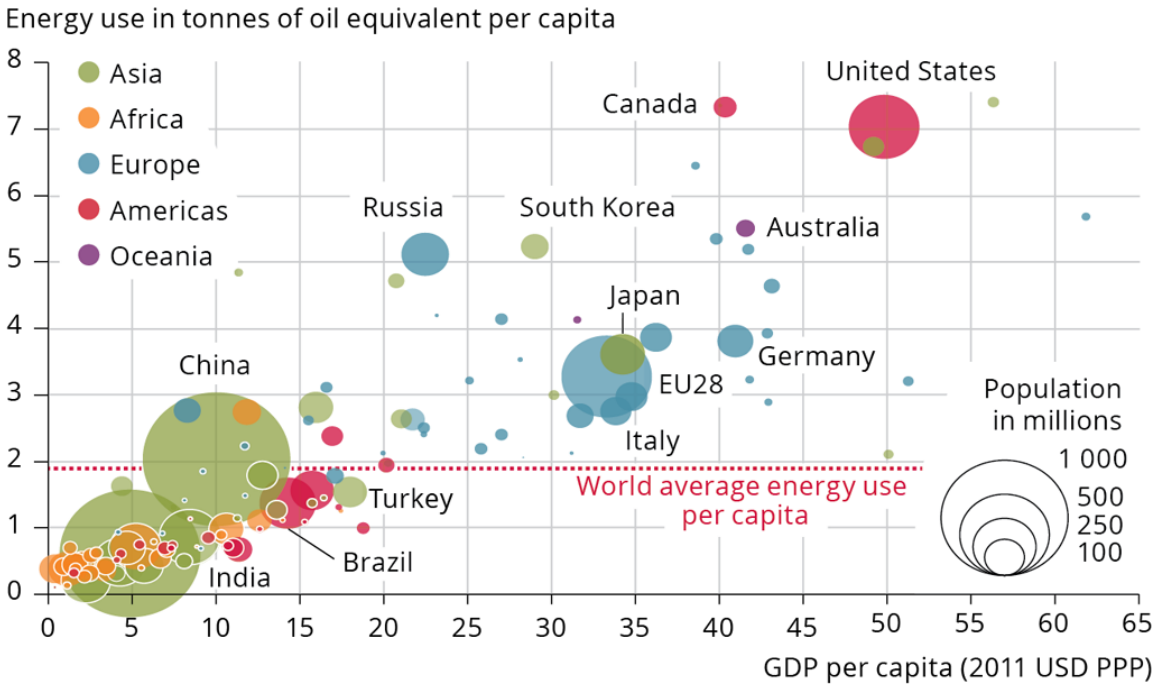
- Climate as a **global** commons
  - Exclusion is difficult.
  - Delaying its depletion is costly.
- Challenges: monitoring compliance and imposing sanctions; spontaneous action without a **global** authority; reducing emissions often increases production costs
  - Failure of the Kyoto Protocol



TRAGEDY OF THE GLOBAL COMMONS

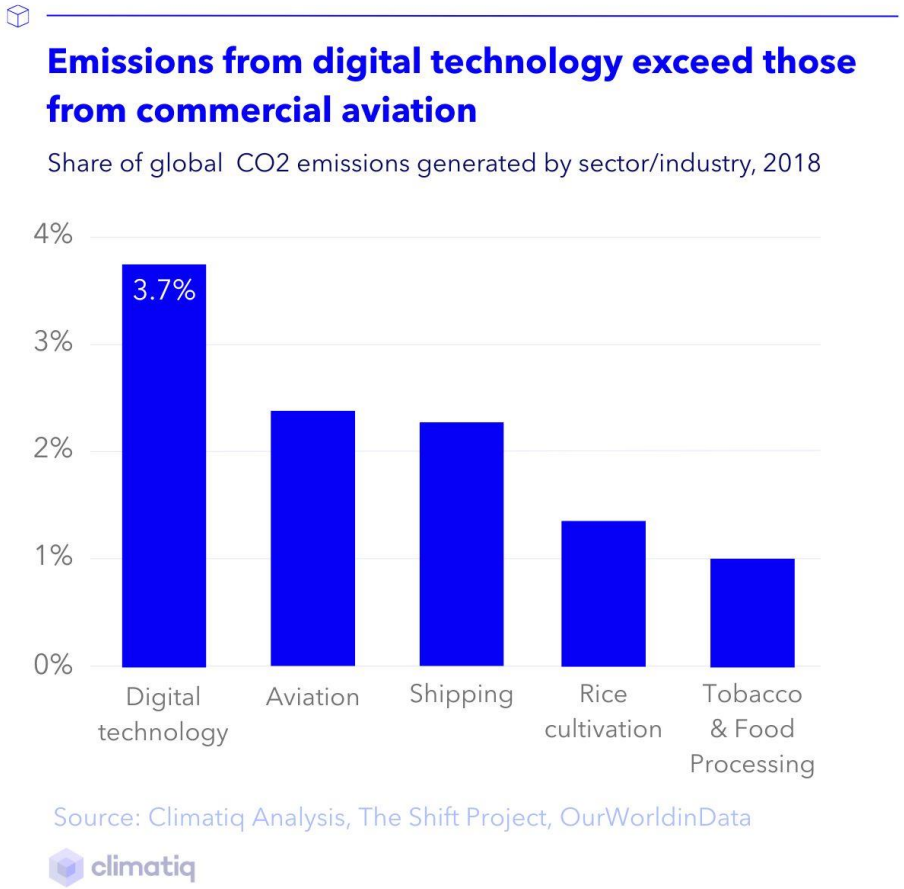
The Energy-Environment Dilemma

- Energy has the confounding characteristic of being both a force of human progress and ecological destruction.
- Economics (**affordability**) and politics (energy **security**)
  - Fossil fuels supply 80% of global primary energy.
  - Energy accessibility in developing countries



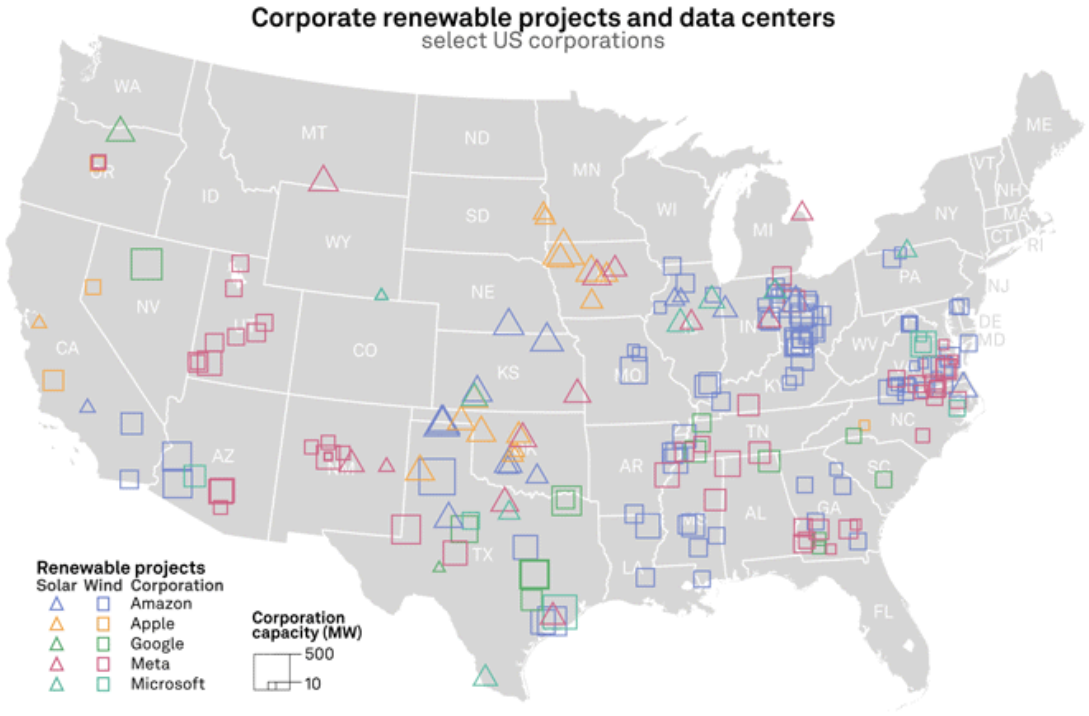
Carbon Contradiction of AI

- Carbon-footprint controversy
- **Scale**-performance coupling: higher capability requires larger models & more data → more energy demand
- **Rebound** effect: efficiency gains in chips/software outpaced by expansion in deployment
- Potential decarbonization benefits vs. risk of fossil lock-in



AI Energy Pathways: The Case of The US

- The AI boom is driving rapid data-center load growth
- AI demand can reinforce short-term **carbon** dependence
- **Renewables** through long-term purchase agreements
- AI-led **nuclear** energy renaissance: AI load invoked to justify nuclear expansion



As of Feb. 1, 2023.
Map credit: Ciaralou Agpalo Palicpic.
Sources: S&P Global Market Intelligence; 451 Research.
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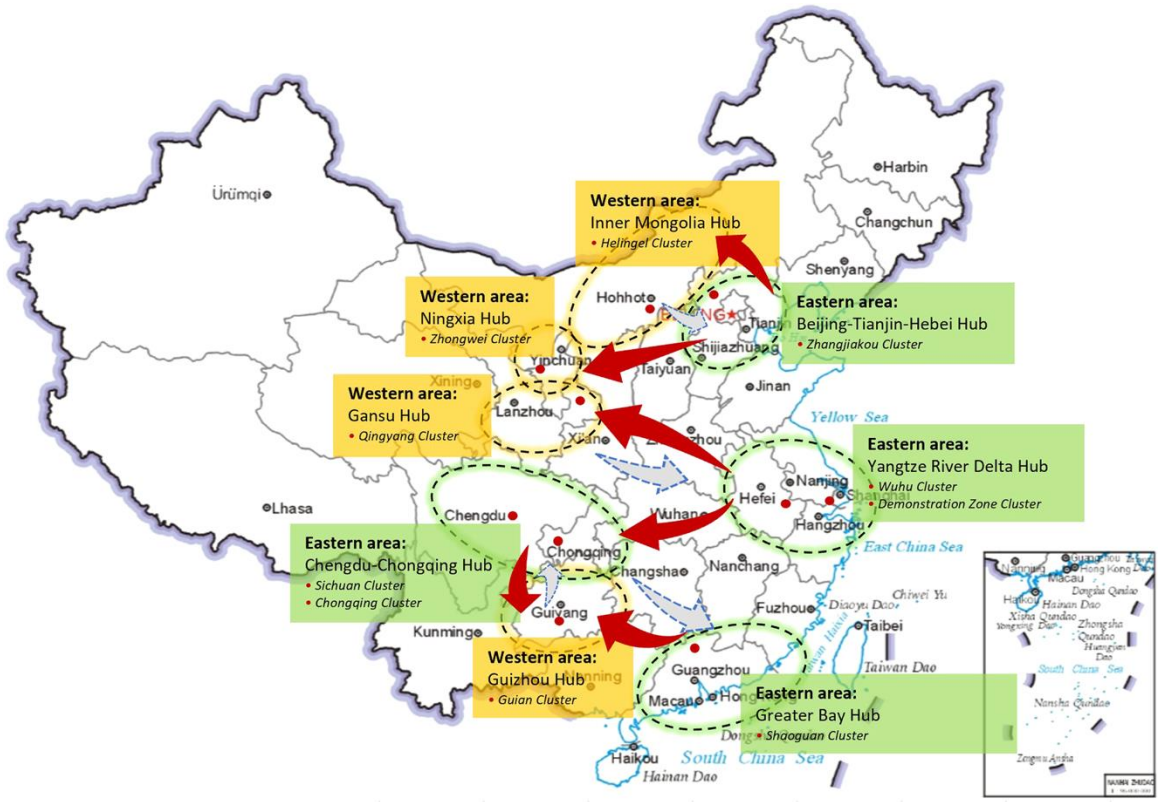
AI Energy Pathways: The Case of China

- State-coordinated energy transition that links data centers, grid expansion, and **renewable** manufacturing
- East data, west computing (东数西算): shift data centers from densely-populated and economically-advanced eastern regions to resource-rich western regions

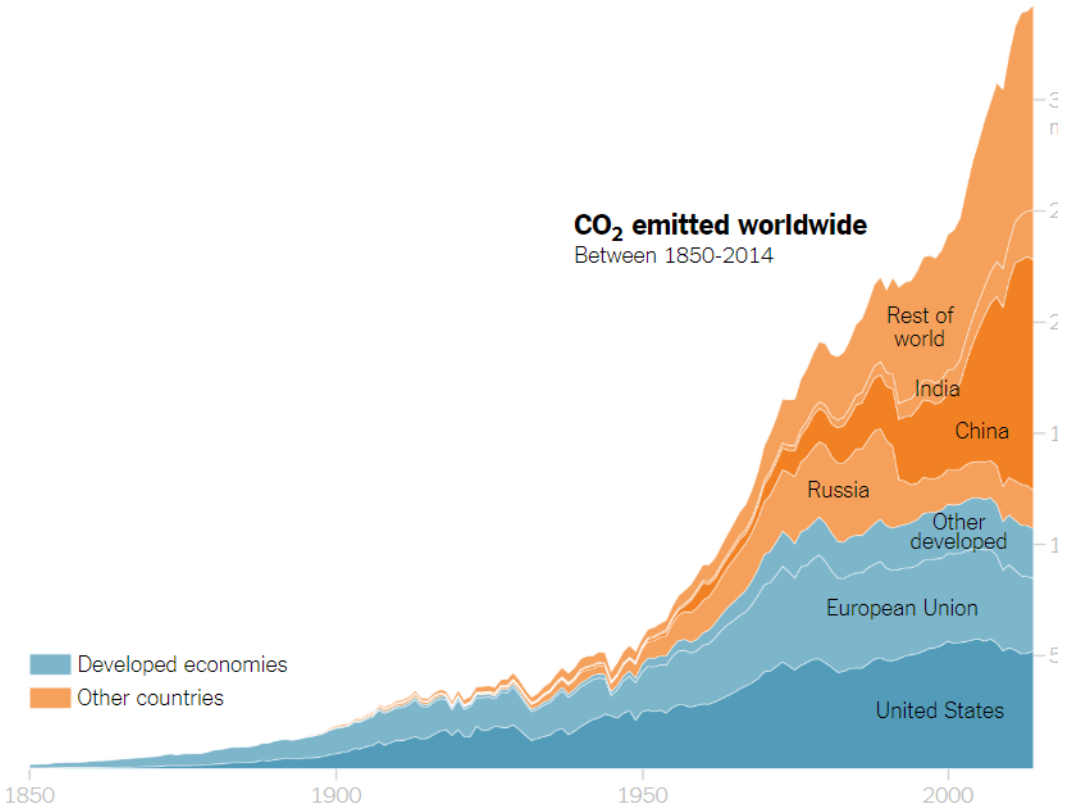


Climate Change

- Climate is the envelope for **all** environmental processes.
- Severe, pervasive, and irreversible impacts
  - Natural: rising sea levels, droughts, storms, flooding, and declining biodiversity
  - Social: food and water scarcity, housing, health, and poverty

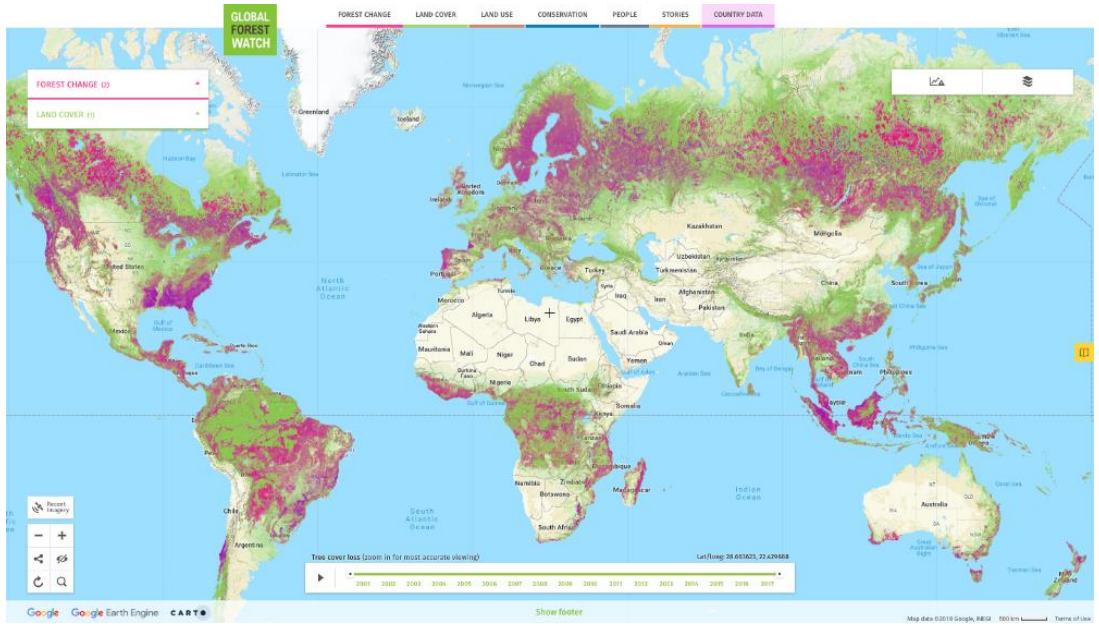
Climate Justice

- Climate impacts are not **created** or **distributed** equally.
  - Unequal history
  - Rich vs. poor, older vs. younger generations are affected differently
- Exacerbating existing inequalities
  - Climate change disrupts food supplies
  - Rising seas force relocation → **climate refugees**



AI for Climate Mitigation

- Climate risk modeling for infrastructure performance
- Smart grids and load forecasting
- Extreme weather prediction
- Deforestation detection using satellite data
- Biodiversity tracking and species detection



Global Forest Watch